

Aviral Garg

📍 New Delhi, India | 📞 +91-9971195728 | ✉️ gargaviral99@gmail.com
🌐 [LinkedIn](#) | 🐙 [GitHub](#)

Professional Summary

Motivated Computer Science student specializing in IIOT with proven skills in cybersecurity, AI/ML, and software development. Published researcher on Intrusion Detection Systems (IDS) for DRDO, known for problem-solving, leadership, and a commitment to delivering high-quality solutions.

Education

University School of Automation and Robotics, GGSIPU, New Delhi

B.Tech in Computer Science (IIOT)

2023 – 2027

Bal Bharati Public School, New Delhi

Senior Secondary Education | 88%

Secondary Education | 92.7%

Experience

AI/ML Intern - Heimatverse

Developing and optimizing AI/ML models to enhance data analysis capabilities.

Research Intern - DRDO (Govt. of India)

Conducted research on Intrusion Detection Systems (IDS); contributed to published research improving IDS methodologies for defense.

Intern - Quine (London-Based ML Development Company)

Contributor - GSSOC 2024 | SSOC 24 | SWOC 25

Organizing Team - Social Summer of Code 2024

Member - Microsoft Startups Founder's Hub

Projects

- **Network Layering Security**
Creating an Application Layer Protocol for tracking Load and Load Balancing; Selected in final round for IPU-IIF Ideathon
- **Prosthetic Arm for Disabled**
Research Project; Using a whole new approach using machine learning for individual control in prosthetic arm; mentorship and guidance by professors at IPU and IIT Delhi.
- **Intrusion Detection System**
Research paper project for DRDO. Created a Machine Learning Model for prediction intrusion for ARP-MiTM and SSDP Floods; Used Multiple Algos; Paper to be Published this January.
- **Speech-to-Speech Translation**
Developing an application for a client project that involves Real Time Speech-to-Speech Translation on Call; Target to reach a latency of less than 0.5 secs using various technologies; Ongoing Research.
- **SocialSellers** | <https://social-sellers.vercel.app/>
Developed a Pipelining Website that uses finely tuned Gemini for Amazon Listing Structure and Multiple APIs for entire website functioning to allow scraping data from social media posts and create Amazon listings; achieved top 25 in Amazon Smbhav Hackathon among 1,00,000+ teams.
- **Money Mantra - Finance Advisory** | <https://money-mantra-pi.vercel.app/>
Developed a finance advisory tool precisely a LLM using finely tuned ChatGPT for finance-based outputs and Azure as Cloud Service for storage and deployment; achieved top 12 in Bank of Baroda & Microsoft Azure Hackathon among 5,000+ teams.
- **Autumn8**
Home Automation Project for the Startup; Working on Recommendation Models and a Voice Assistant just like TARS/Jarvis (To be launched in production next month).
- **Codebot** | <https://github.com/aviralgarg05/Codebot>
Automated PR review GitHub bot with Fine Tuned LLM via MindsDB using RAG chains; awarded in Quira Quest-14.
- **MindMate** | <https://github.com/aviralgarg05/MindMate>
Mental health assistant using Fined Tuned OpenAI, Twilio, Google Maps APIs, and anti-suicidal support; awarded in Quira Quest.
- **Stock Market Prediction System** | <https://github.com/aviralgarg05/Basic-Stock-Price-Prediction>
Machine Learning models and algorithms for stock prediction. Further work includes construction of Algo Trading bot; selected in top 1000 projects at DevFolio Quest-10.

- **Twitter Sentiment Analysis** | <https://github.com/aviralgarg05/Twitter-Sentiment-Analysis>
Sentiment analysis tool using NLP libraries to filter hateful/abusive content.
- **Productivity Measurement System for Corporates** | <https://github.com/aviralgarg05/Face-Emotion-Recognition>
Developed a system to track and analyze employee productivity using OpenCV and DeepFace, aiding workforce optimization.
- **Plant Disease Diagnostics Tool** | <https://github.com/aviralgarg05/Leaf-disease-detection>
ML-based tool for plant disease detection using image analysis using OpenCV.
- **Loan Repayment Prediction**
AI model predicting loan repayment ability based on financial data using LSTMs and Bi-LSTMs; Part of Open Source Contribution.
- **CoraBand**
COVID-19 prevention device; awarded by Young Scientists India with ISRO collaboration.

Skills

Languages & Frameworks: Python, C, C++, JavaScript, HTML, CSS, Express, FAST, FLASK

Data & ML Libraries: TensorFlow, Keras, Pandas, Numpy, Scikit-Learn, OpenCV, NLP, Wave2Vec

Tools: Azure Cloud, Google Cloud, Docker, Postman, API Handling, Git, GitHub Actions, Postgres

Database Management: MySQL, MariaDB, MindsDB, MongoDB

Version Control: Git, GitHub

Languages: English, Hindi